EWB CYBERSECURITY
ASSESSMENT 01

Name: Mohammad Yasir

Course: AI-Powered Cybersecurity Internship

Organisation: EWB Courses

Tasks included

Task A — Secure a Student Account
Task B — Analyse a Suspicious Message
Task C — Triage a Sign-in Alert
Task D — Review an AI Security Answer

Task A — Secure a Student Account

A1. Risks, assets, CIA impact, controls and limitations

Mira's habits can put her accounts, laptop and project files at risk. The table lists five different risks and a practical control for each one.

Risk	Asset	CIA impact	Control	Limitation
Reusing a short password	Email and learning portal	Confidentiality: if the password is exposed, someone may try it on both accounts.	Use a long, unique password for each account. A password manager can help.	A strong password can still be stolen through phishing.
Approving unexpected sign-in prompts	Learning portal account	Confidentiality and integrity: another person may access or change information.	Deny prompts Mira did not start and report repeated requests.	MFA is useful, but it does not stop every account attack.
Postponing device updates	Laptop and stored information	Confidentiality, integrity and availability may be affected by known weaknesses.	Install security updates regularly and restart when needed.	Updates cannot protect against every weakness or new attack.
Keeping only one project copy	Project files	Availability: files may be lost if the laptop fails, is damaged or goes missing.	Keep a second copy on separate storage or a trusted cloud service; check it can be restored.	An old or compromised backup may not help.
Sharing a laptop while signed in	Personal files and learning portal	Confidentiality and integrity: another user may view files or make changes.	Lock the laptop and create a separate user account for the friend.	Separate accounts reduce risk but do not remove every risk from physical access.

A2. My first three priorities

The first thing I would do is change the reused password so they are different. I would do this because multiple accounts can be affected by a single leaked password. I would then no longer approve any unexpected MFA alerts. This is important because it can allow another person to login. Finally I would lock my laptop and create a new account to use on the laptop. This will help keep Mira's files and learning portal secure. Finally I will install any updates that are waiting to be installed, along with creating a new backup of the project files. The first 3 steps would help reduce any risk of account and access issues.

A3. Authentication and authorisation

Authentication is identifying who you are. This can be done by Mira entering her password and completing MFA to prove she is the owner of the account. Once you have authenticated you are authorised to do certain tasks. For example Mira would only be able to view the content of the course and submit assignments but would not be able to edit settings which are for the college administrator. Authentication is proving you are who you say you are whilst authorisation gives you permission to do certain tasks.

Task B — Analyse a Suspicious Message

B1. Five warning signs

I would treat this message as suspicious and verify it separately. These are warning signs in the supplied fictional message; they do not by themselves prove who sent it.

Evidence from message	Why it needs scrutiny
“awards@campus-awards.example”	The sender domain is different from the college's normal campus.example domain.
“Reply-To: help@personal-awards.example”	The reply address differs from the sender address. Replies may go somewhere unexpected.
“scholarship expires in 20 minutes”	This creates pressure and may make the student act without checking.
“enter your college password and OTP”	An unexpected message should not ask the student to provide login secrets.
“Do not contact the college office”	It discourages the student from checking the claim through an independent official channel.

B2. Safe verification process

I would not click the link or reply to the message. I would open the college portal by typing its known address myself, or contact the college office using a phone number or email obtained separately from an official source. I would ask whether the scholarship and request are genuine. HTTPS only means the connection is encrypted; it does not prove that the website belongs to the college. A familiar display name can also be copied, so I would check the actual sender details and confirm through an official channel.

B3. If the student already entered a password and OTP

1. Contact the college's official IT support immediately. Do not send the password or OTP to anyone.
2. Change the affected password using the official portal. If it was reused elsewhere, change it there too.
3. Ask IT to revoke active sessions and check recent account activity.
4. Follow IT's instructions to secure or reset MFA. Do not approve more unexpected prompts.
5. Keep the suspicious message as evidence and report it through the college's official process.

B4. What cannot be established

From this message alone, I cannot confirm the real identity of the sender or who controls the linked website. More evidence could include the full email headers, confirmation from the college through an official contact, and a technical review of the website by authorised staff. The text is enough to justify caution, but not enough to identify a person.

Task C — Triage a Sign-in Alert

C1. Four-point timeline

The times below are from the synthetic log supplied in the assessment. I have separated what the log shows from what I think it may mean.

Time	Observed event	My interpretation
09:00–09:01	Password success and MFA approved from 192.0.2.10.	This matches the usual classroom egress stated in the lab notes, so it looks like the normal sign-in.
10:14–10:15	Two password failures from 198.51.100.23.	Repeated failures are suspicious and may be attempts to guess or use a password.
10:16–10:18	Password success, MFA denied, then MFA approved from 198.51.100.23.	The password was accepted and the MFA prompt was later approved. The student says they approved it to stop repeated prompts.
10:20	A session was created from 198.51.100.23.	A session followed the MFA approval. This needs investigation because the student says they did not sign in.

C2. Likely explanation, alternative and checks

A likely explanation is that someone had obtained the student's password and tried to sign in. After two failed attempts, the password worked. The MFA request was first denied and then approved by the student, who says they thought approval would stop the repeated requests. This may have allowed the session to be created.

An alternative explanation is that the sign-in came from a legitimate device or network that had a different IP address. A different IP address alone does not prove an attacker was involved.

Two useful checks are: (1) review device, browser, session and location information available to the authorised IT team; and (2) compare MFA records and ask the student whether they recognise the device or activity.

C3. Three prioritised actions for the authorised IT team

1. Preserve the sign-in logs, MFA records and session details before making changes, so useful evidence is not lost.
2. Protect the account by revoking suspicious sessions and arranging a secure password reset. Check whether other accounts use the same password.
3. Contact the student through an official channel, verify the activity and help secure MFA. Continue checking for further unusual activity.

C4. Incident summary (60–100 words)

The logs show two failed password attempts followed by a successful password entry from an unfamiliar IP address. MFA was denied and then approved, and a session was created at 10:20. The student says they did not sign in and approved the prompt because they wanted the repeated requests to stop. This suggests possible unauthorised access, but the IP address does not identify a person. The IT team should preserve the evidence, secure the account and investigate the device and MFA records.

Task D — Review an AI Security Answer

D1. Five unsafe or unsupported claims/actions and corrections

AI output	Correction
“The email is safe because it uses HTTPS.”	HTTPS encrypts a connection but does not prove that the sender or website is genuine. The message still has warning signs.
“The sign-in logs prove that the person using 198.51.100.23 stole the password.”	The logs show events from an IP address. They do not identify a person or prove how the password was obtained.
“MFA prevents every account compromise.”	MFA adds protection, but a user can still be tricked into approving an unexpected prompt. The student says this happened in the scenario.
“Delete the suspicious logs.”	The logs should be preserved because they may help the authorised IT team understand what happened.
“Send the student's password to support so they can test it.”	Passwords and OTPs should not be shared. Support should use a secure account recovery process.

D2. A better prompt

“Please analyse only the fictional email and sign-in log evidence provided. Separate observed facts from assumptions. Identify possible security risks, explain what cannot be confirmed, and suggest safe response actions. Do not request or include passwords, OTPs, identity documents or real personal information. If information is missing, clearly say what additional evidence would be needed.”

D3. My corrected security advice (100–150 words)

A suspicious message should be checked before taking action. Do not trust an email only because it uses HTTPS or shows a familiar name. Never share your password, OTP or identity documents through an unexpected message. Verify requests through the official college portal or a trusted contact found separately. Repeated sign-in requests should not be approved unless you started them. If you approve an unexpected request or share your login details, contact the official IT team immediately, change your password and ask them to revoke suspicious sessions. Security logs should be kept for investigation. MFA, strong passwords and regular updates improve account security, but no single security control can guarantee complete protection.

D4. AI-use declaration

AI-use declaration: ChatGPT was used to help organise the assessment answers, explain some cybersecurity concepts and improve clarity. I reviewed the answers against the fictional evidence in the assessment and am responsible for checking and understanding the final submission.

References and evidence notes

1. EWB Courses, Assessment 01: Security Foundations, issued 18 October 2026. Tasks A–D and the fictional scenarios/logs supplied in the assessment.
2. CISA, Secure Our World: https://www.cisa.gov/secure-our-world
3. NIST, Generative AI Profile, AI 600-1: https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf